

Towards a Reproducible Spatio-Temporal Transportation Network Analysis Toolkit

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Abstract

The growing availability of urban mobility data, especially standardized sources such as the General Transit Feed Specification (GTFS) and collaboratively maintained resources like OpenStreetMap (OSM), has enabled increasingly fine-grained assessments of public transit accessibility across space and time. Despite this, researchers often construct bespoke workflows to combine GTFS, OSM, and demographic datasets (e.g., the American Community Survey, ACS). These ad hoc pipelines frequently limit reproducibility, require repeated data integration efforts, and introduce unnecessary friction when extending analyses across multiple time periods or regions.

This work introduces *R5-ST*, a Python-based processing framework designed to streamline the full end-to-end workflow of spatio-temporal transit accessibility analysis. Built around the R5 routing engine and its Python interface *r5py*, R5-ST automates the generation of origin–destination travel time matrices and enables structured comparisons of transit travel metrics across years or decades of archived GTFS data. The framework also includes native support for joining travel time outputs with demographic attributes, allowing researchers to compute accessibility metrics conditioned on population characteristics. By consolidating these tasks, R5-ST reduces redundant preprocessing steps and facilitates consistent, replicable analyses of transit system performance and its contribution to urban accessibility.

R5-ST is generalized for use in any urban region for which GTFS and OpenStreetMap (OSM) data are available. Users can specify GTFS and OSM inputs, origin sets, and points of interest (POIs), which may either be provided directly by the researcher or retrieved automatically from OSM. The framework supports user-defined temporal sampling intervals (e.g., weekly), time windows of interest, and multimodal travel configurations (e.g., walk, transit, car). When demographic data such as ACS variables are available, these can be incorporated to enable integrated equity-focused analyses. The framework also natively supports execution within high-performance computing environments, facilitating large-scale or multi-year studies across extensive sets of origins and destinations.

To demonstrate the framework, we apply R5-ST to the Boston metropolitan area using the Massachusetts Bay Transportation Authority (MBTA) GTFS feeds from 2019–2025. The analysis considers all census block groups as origins and all K–12 schools in the region as destinations. Weekly travel time matrices were generated over the six-year period, and demographic analysis was performed on the resulting time series to identify how accessibility for different population groups evolved over time.

By unifying data ingestion, computation, compilation, and demographic evaluation, R5-ST provides a reproducible and extensible foundation for transit accessibility research. Potential applications include monitoring accessibility trends over time, assessing equity impacts of service changes, and examining access to key destinations such as schools, healthcare facilities, and commercial centers.